

Pierre Moreau

PhD in Computer Science · Data Scientist / ML Engineer — AI & Healthcare

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Summary

PhD in Computer Science (AI / Machine Learning) specialised in healthcare, combining applied research and engineering. Skilled in multimodal learning, generative AI (LLMs, RAG) and clinical data processing. Committed to building interpretable, reliable AI for real-world clinical impact.

Skills

Languages: Python, SQL, JavaScript, Bash, PHP, HTML/CSS

ML & Deep Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers, scikit-learn, XGBoost

LLMs & Generative AI: Fine-tuning (PEFT, LoRA), RAG (LangChain), prompting, Whisper (speech-to-text), OCR

Vision & NLP: Segmentation (UNetR, SwinUNet), OpenCV, Pillow, NLTK

Data Science: NumPy, Pandas, SciPy, Matplotlib, Seaborn ; PCA, DTW, ANOVA, Chi²

MLOps & Deployment: FastAPI, Django, Flask, Docker, Git

Responsible AI: Explainability (SHAP, LIME), bias/fairness analysis, GDPR & AI Act

Tools: Git, Docker, VS Code / Cursor, Jupyter, Claude Code / Codex

Professional Experience

Post-doctoral Researcher — AI & Healthcare, University of Picardie Jules Verne – Amiens Sept 2025 – present

- RAUC project: designing an optimised care pathway for digestive surgical emergencies, integrating AI and e-health, aimed at predicting post-operative complications.

Data & AI Engineer — AI & Data Science Lead (Healthcare), Capgemini Engineering – Paris Mar 2024 – May 2025

- Designed an AI-based medical assistant demonstrator with multimodal workflows.
- Built NLP pipelines: OCR of blood test results, speech-to-text transcription (Whisper), automatic keyword extraction to enrich metadata.
- Fine-tuned LLMs (OpenAI, Mistral) with Hugging Face Transformers and PEFT for clinical summaries and audio transcription.
- Developed a multimodal RAG pipeline (LangChain) synthesising structured and unstructured medical data.
- Applied research on placenta segmentation (UNetR, SwinUNet) to improve accuracy in prenatal imaging.
- Coordinated AI researchers, data scientists and clinicians; project and team management (15-20 people).

Teaching & Research Assistant (ATER), University of Picardie Jules Verne – Amiens Sept 2022 – Aug 2023

- Tutorials and lab sessions for undergraduates (years 1-3) — PHP, Bash, C, advanced programming.

PhD Candidate — Doctoral Research, University of Picardie Jules Verne – Amiens Sept 2019 – June 2023

- Research on human motion modelling and analysis (in collaboration with Amiens University Hospital); built a comprehensive motion database.
- Analysis (PCA, Dynamic Time Warping, correlation) and optimisation via genetic algorithms for feature selection.
- Deep learning modelling (RNN, LSTM, CNN, residual networks, FcNN) and SVM; interpretable biomechanical representation using matrix formulations.

IT Engineer / Developer (apprenticeship then permanent), LV Informatique – Amiens Sept 2017 – Sept 2019

- Developed custom ERP software (Lynéade then Lynéa2, WinDev): business processes (accounting, purchasing, sales, inventory, after-sales), SQL database administration, client support and customisation.

Education

University of Picardie Jules Verne, PhD in Computer Science — Artificial Intelligence – Amiens Sept 2019 – June 2023

- Thesis: analysis of human motion using artificial intelligence tools, in collaboration with Amiens University Hospital.

University of Picardie Jules Verne, MSc in Computer Methods Applied to Business Management (MIAGE) – Amiens Sept 2017 – June 2019

Publications

- A Genetic Algorithm-Optimized PID Controller for Robust Tumor Treatment via Combined Immuno-Chemotherapy** 2026
Jérôme Bosche, Consibelle Djiffouo Dzoyim, *Pierre Moreau*
(International Conference on Control, Decision and Information Technologies (CoDIT) — submitted)
- Towards Better Placenta Segmentation for Earlier Preeclampsia Detection on 3D Doppler Images** 2025
Charlotte Alliod, Emilien Micard, *Pierre Moreau*, Léo Challier, et al.
(Computerized Medical Imaging and Graphics (CMIG) — submitted)
- Two-branch neural network using two data types for human activity recognition** 2023
Pierre Moreau, David Durand, Jérôme Bosche, Michel Lefranc
(IEEE Sensors Journal)
- A motion recognition technique based on linear matrix representation to improve Parkinson's disease treatments** 2021
Pierre Moreau, David Durand, Jérôme Bosche, Michel Lefranc
(9th International Conference on Systems and Control (ICSC))
- A motion recognition algorithm using polytopic modeling** 2020
Pierre Moreau, David Durand, Jérôme Bosche, Michel Lefranc
(7th International Conference on Control, Decision and Information Technologies (CoDIT))

Certifications

- Generative AI with Large Language Models — DeepLearning.AI & AWS (2024)
Building Multimodal Search and RAG — DeepLearning.AI (2025)
Build Basic Generative Adversarial Network (GAN) — Coursera (2025)
Python NLTK for Beginners — Coursera (2024)